

Week 6 Sean Graham

1.

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

loan x Untitled1* x

Filter

	id	loan_amnt	term	int_rate	installment	grade	emp_length	home_ownership
1	1077501	5000	36 months	10.65	162.87	B	10+ years	RENT
2	1077430	2500	60 months	15.27	59.83	C	< 1 year	RENT
3	1077175	2400	36 months	15.96	84.33	C	10+ years	RENT
4	1076863	10000	36 months	13.49	339.31	C	10+ years	RENT
5	1075358	3000	60 months	12.69	67.79	B	1 year	RENT
6	1075269	5000	36 months	7.90	156.46	A	3 years	RENT
7	1069639	7000	60 months	15.96	170.08	C	8 years	RENT
8	1072053	3000	36 months	16.64	109.43	E	9 years	RENT
9	1071795	5600	60 months	21.28	152.39	F	4 years	OWN
10	1071570	5375	60 months	12.69	121.45	B	< 1 year	RENT
11	1070078	6500	60 months	14.65	153.45	C	5 years	OWN
12	1069908	12000	36 months	12.69	402.54	B	10+ years	OWN
13	1064687	9000	36 months	13.49	305.38	C	< 1 year	RENT

Showing 1 to 15 of 887,379 entries, 11 total columns

Console Terminal Jobs

```
[workspace loaded from ~/.RData]
> library(readxl)
> loan <- read_excel("loan.xlsx")
> View(loan)
> |
```

2.

RStudio

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Go to file/function Addins

Project: (None)

loan x Untitled1* x

Source on Save Run Source

```
1 library(readxl)
2 loan <- read_excel("loan.xlsx")
3 View(loan)
4
5 #2 Plot the histogram and density of loan_amnt and add mean line using ggplot2
6 library(ggplot2)
7 ggplot(data = loan, aes(x=loan_amnt))+
8   geom_histogram(aes(y=..density..), color="black", fill="white")+
9   geom_density(alpha=.2, color="red", size=1)+
10  geom_vline(aes(xintercept=mean(loan_amnt)), color="blue", size=1)
```

Environment History Connections

Global Environment

Data

loan 887379 obs. of 11 variables

values

	my.cols	chr [1:2]	"green"	"blue"
tbl.term	'xtabs'	int [1:2, 1:7]	143015	5187 212...
terms	Factor w/ 2 levels	"36 months", "60 mon...		

Files Plots Packages Help Viewer

Zoom Export

density

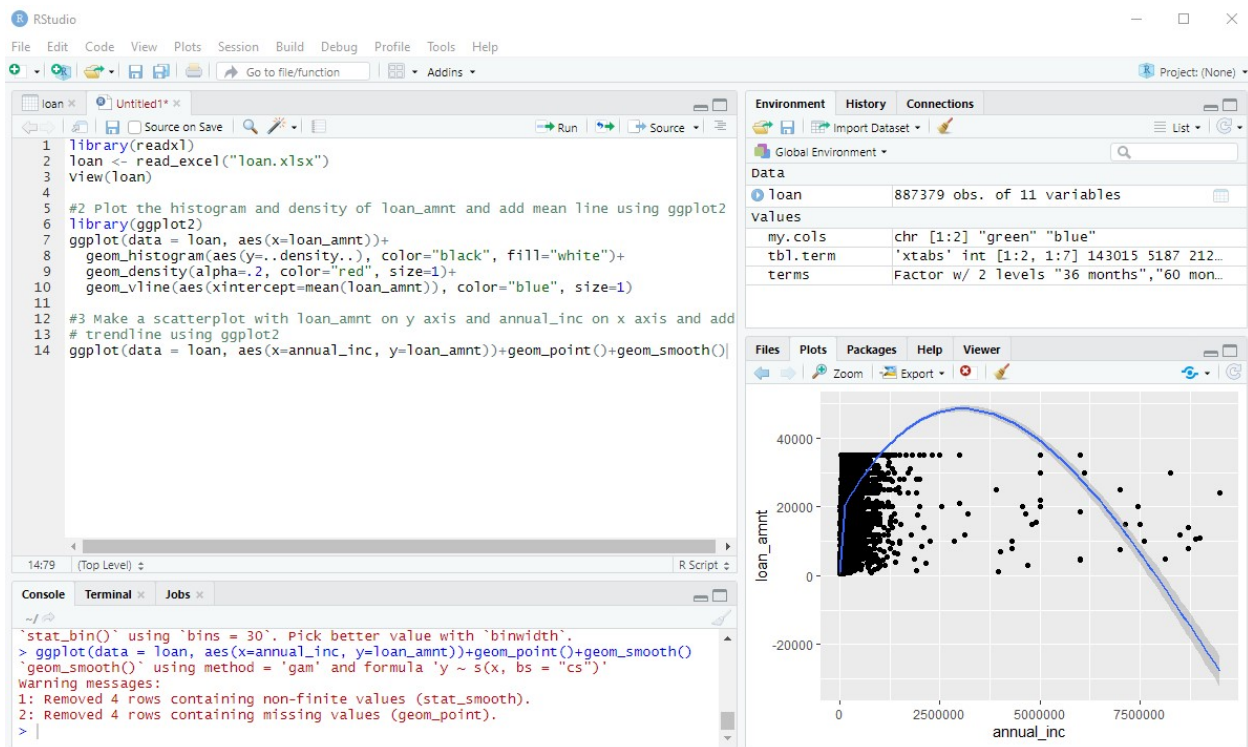
loan_amnt

1:16 (Top Level) R Script

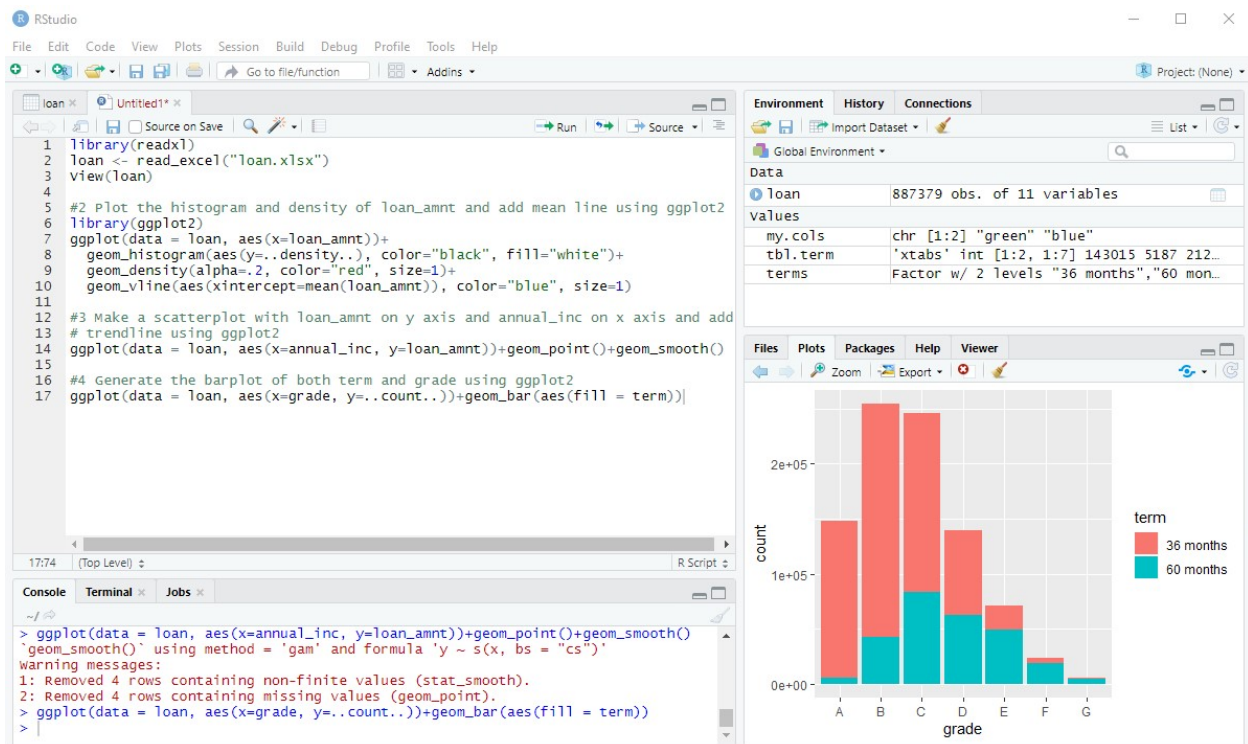
Console Terminal Jobs

```
> library(ggplot2)
> ggplot(data = loan, aes(x=loan_amnt))+
+   geom_histogram(aes(y=..density..), color="black", fill="white")+
+   geom_density(alpha=.2, color="red", size=1)+
+   geom_vline(aes(xintercept=mean(loan_amnt)), color="blue", size=1)
'geom_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
> |
```

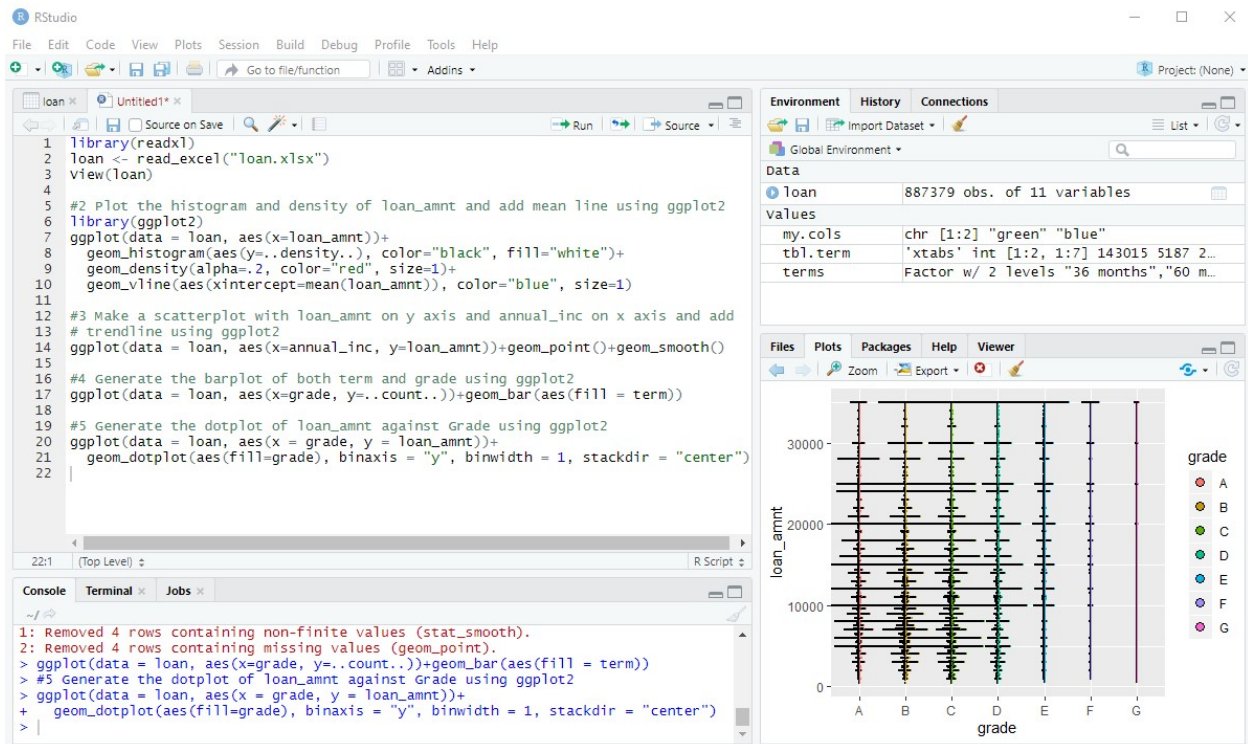
3.



4.



5.



6. Loans with terms of 60 months tend to be larger than those with terms of 36 months. The notches are very narrow, but they're still there, and they don't overlap, so we can say with 95% confidence that the medians are different.

